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Effects of simulation-based education on nursing students' perceptions of positive birth and natural birth approaches[☆]

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KEYWORDS

Birth care;
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Natural birth;
Nursing education;
Nursing student;
Simulation

Abstract

Background: Nursing education should be adjusted to align with national health priorities. In this study, we examined how simulation-based education impacted nursing students' knowledge, skills, and approaches to natural childbirth, as well as their self-confidence, perceptions of positive birth, and educational satisfaction.

Methods: A total of 56 undergraduate nursing students were enrolled in this comparative, quasi-experimental, pre- and post-test designed study and were randomly assigned to a hybrid simulation (HS) or a video-model (VM) group.

Results: Students in the HS group had significantly higher natural childbirth skill levels and positive perceptions than those in the VM group ($p < .05$). Natural birth knowledge levels significantly increased in both groups ($p < .05$). No significant differences in nursing students' self-confidence and satisfaction levels were found between the two groups ($p > .05$).

Conclusion: HS improved nursing students' natural birth approach skills and positively affected their perceptions. Simulation-based education may enhance maternity care practice. Simulation scenarios should be integrated into the nursing curriculum to provide comprehensive support for positive birth experiences.

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Introduction

Natural birth is a delivery where the mother solely relies on her instincts and takes an active role without medical intervention (Rathfisch, 2018). Giving birth is a natural and unique experience for every woman. In general, childbirth occurs without any complications for the majority of women and their babies. Allowing this process without medical intervention is critical in preventing a negative birth experience (WHO, 2018a). The prevalence of negative birth experiences among women ranges from 6.8% to 44%, associated with physical and psychological consequences for both the mother and family (Bell & Andersson, 2016; Dencker et al., 2019). The World Health Organization (WHO) emphasizes the importance of respectful treatment, effective communication, support, and continuity of care throughout labor for a positive birth experience (WHO, 2018b). Health professionals are key stakeholders in providing care that aligns with these factors. Professional attitudes and competence of healthcare providers and trusting caregivers' skills during labor are crucial for a positive birth experience (Hall et al., 2023; Yousefi-Ghalehroodkhani et al., 2024). Although there is no reported rate of negative birth experiences in Türkiye, the factors associated with this situation are high. The WHO recommends that the cesarean section rate should be between 10% and 15% and should only be used in situations where vaginal delivery poses a risk to the mother or baby (Curran et al., 2016; Turkish Perinatal Society, 2019). The latest data from the Organization for Economic Cooperation and Development (OECD) in 2020 indicated that Türkiye had the highest cesarean section (CS) rate among its member countries. The national data also reported that the CS rate was 57.55% (n = 6,161,976) between 2018 and 2023 (Ulgu et al., 2023). Furthermore, the rate of fear of childbirth among pregnant women in the country has been reportedly higher than that in Western countries (OECD, 2024; Kahraman & Gökçe İsbir, 2023).

Factors such as minimal intervention, natural relaxation options, a positive birth environment, and birth support, which are at the core of natural birth, can make a positive birth experience more possible. Studies have shown that nurses and midwives working in obstetrics clinics lack knowledge about natural childbirth. Moreover, nega-

tive attitudes and perceptions of healthcare professionals create barriers to high-quality birth care. In this phase, training midwives and nurses on natural childbirth can enhance women's desire for a natural birth (Hall et al., 2023; Yousefi-Ghalehroodkhani et al., 2024).

Since 2013, the Turkish Ministry of Health has been setting targets in line with that of the WHO to improve the disadvantageous situation. One of the goals at the clinical level is for health professionals to develop positive attitudes and behaviors toward natural birth and to receive training on positive birth experiences (Turkish Society of Obstetrics and Gynecology, 2013; Güleç Şatır et al., 2018; Yousefi-Ghalehroodkhani et al., 2024). The WHO recommends evaluating nursing education curricula and practices in the context of educational needs and national health priorities. Training competent healthcare professionals with evidence-based, technology-first teaching methods facilitating active learning is the foundation for achieving learning outcomes (WHO, 2016).

The medicalization of birth and high rates of cesarean sections prevent nursing students from providing natural birth care. Effective educational strategies should be investigated to teach nursing students about this issue and disseminate the findings (Birkhead et al., 2012). Simulation-based education (SBE) has become increasingly well-known in nursing education, providing students with a safe and controlled environment to practice their skills (McDonald et al., 2021). Previous studies reported that SBE positively impacts students' self-confidence, satisfaction, and medical assessment results. Learning vaginal delivery skills was more effective through simulations than in a clinical setting. Obstetric simulations have been proven to improve psychomotor skills and establish a connection with evidence-based practices, offering motivation (Xu et al., 2021; Pajohideh et al., 2023; Leal-Costa et al., 2024). Furthermore, research on the relationship between SBE and natural birth approaches is limited (Nugraha et al., 2018). This study aims to address these research gaps.

Aim

This study aimed to assess the impact of SBE developed based on evidence-based practices outlined in the WHO

and International Lamaze Organization guidelines, on nursing students' attitudes toward natural birth approaches and positive birth experiences. The research questions were as follows:

1. Does using hybrid simulation affect the natural birth knowledge?
2. Does using hybrid simulation affect the natural birth approach?
3. Does using hybrid simulation affect students' satisfaction and self-confidence?
4. Does using hybrid simulation affect the natural birth approach skills levels?

Material and methods

A randomized quasi-experimental design was used, and a pre- and post-test survey was conducted. Data were collected from May 2023 to October 2023.

Study population and sample

The study involved 288 undergraduate nursing students enrolled in a nursing faculty in Izmir, Türkiye, during the 2022-2023 academic year. The study was conducted with eighth-semester undergraduate students taking the Women's Health and Diseases course. All nursing students participated in the study, except those working as clinic nurses simultaneously and not undergoing theoretical and video-model training. The researcher invited students to participate in the research by making announcements through their class representatives using the WhatsApp application and by making verbal announcements during the break. Participants were informed that taking part in the research was entirely voluntary. The researcher, who was not actively involved in the participants' education, collected voluntary participant information to avoid exerting authority over them. Additionally, students were informed that whether they chose to participate would not impact their academic standing. As part of the study, the students who volunteered were categorized into three groups based on their high, medium, and low academic success levels. These groups were sorted based on their average success rate in descending order to assess whether their academic performance would affect their knowledge of natural birth. Students were placed in the hybrid simulation (HS) and video-modeling (VM) groups using a simple random sampling technique based on their success levels. At the beginning of the study, students were invited to participate. A confidentiality agreement has been created for students to ensure they do not share any information or communicate with others about the education they receive, and their signatures have been obtained. The study included 56 students, with an equal distribution between the HS and VM groups.

Sample size

The study's sample size was determined based on the educational satisfaction variable. To analyze the effects of the simulation-based training method, satisfaction levels from a study conducted by Bakır and Atan in 2018 were used (Bakır & Ünsal Atan, 2023). The analysis revealed an effect size of 0.70. To detect a significant difference using a Student's t-test for independent samples with a desired statistical power of 0.80 and an alpha significance level of 0.05, a sample size calculation using G*Power 3.0 indicated that 26 participants per group would be necessary (Faul et al., 2007). The study was conducted with 28 students in each group.

Interventions

The HS was designed following the Healthcare Simulation Standards of Best Practices™ (INACSL Standards Committee, 2021). The 2018 Türkiye Demographic and Health Survey (2018 TDHS) data, the WHO evidence-based recommendations on childbirth, and the National Mother-Friendly Hospital Criteria were utilized to determine the requirements (Turkish Ministry of Health, 2024). The SMART framework was used to establish a measurable goal for the simulation (Doran, 1981). Figure 1 shows the HS steps (Cheng et al., 2016). In the HS, the standardized patient portrayed the role of the pregnant woman with low fidelity pregnant abdominal model was utilized in a simulated natural labor room. The SP was a postgraduate student with previous SP experience and training in natural childbirth. After being informed about the study, the SP provided consent and completed four hours of training on natural childbirth and the simulation scenario.

The scenario was divided into ten states (Figure 2). Groups of two students came to the simulation laboratory based on the implementation schedule one week after theoretical training. Practices were performed by acting out scenarios as patient relatives and nurses. Video debriefing sessions followed. The VM group watched a training video that followed the "Natural Approach to Childbirth Scenario" with a standardized patient and a volunteer student. After one week of theoretical training, students in this group were presented with a video in the classroom environment. The video was also sent to them as reference material so they could watch it again before practicing the skill. After the training video, a feedback session was conducted with students to evaluate whether they had learned the goals set for the natural birth approach. The VM group's feedback session lasted for 15–20 minutes. The evaluation phase began two weeks after completing the training for each group.

In the first stage, both groups completed the Natural Birth Approach Questionnaire, Student Satisfaction and Confidence in Learning Scale, and Satisfaction with the

Components	Implementation in HS
Learning objectives	To provide an evidence-based, natural birth approach during the first stage of labor for pregnant women in compliance with the National Mother-Friendly Hospital Criteria
Student Preparation	“Theoretical Training for Developing Positive Birth Perception and Approaching Natural Birth.” Topic: Philosophy of natural birth, Expectation of birth and the importance of the birth experience, Birth knowledge and pushing techniques, and Non-pharmacological methods to cope with labor pain. Time: 200 minutes
Scenario	Melek, who has started labor, has been admitted to room 214, accompanied by a relative. When Mine arrived at the hospital, she was already 3 cm dilated and experiencing pain. Melek previously agreed that she would have a natural delivery, and her labor is expected to progress naturally.
Environment	A simulation room has been designed specifically for natural childbirth. “A hospital bed, four stools, pilates ball, mat, rope hanging from the ceiling, hot water bag, candle, source of light, incense, laptop for music, blood pressure monitor for vital signs, thermometer, heat radiating fan, water pool, screens and room decorations (balloons, paper decorations, etc.), NST device.”
Facilitators	Proficient in simulation-based nurse educators Nurse educators in women's health and diseases A checklist with 21 items for evaluating care based on the natural birth approach was developed by researchers. A guide showing the flow of the natural birth approach scenario during its implementation
Pre-Briefing	Orientation to the environment was provided, and expectations regarding the scenario were shared. Requirements were explained before the simulation, and the roles of all participants were defined and shared with the students.
Debriefing	Plus delta method (reaction, description, and summarization) by the researcher nursing educators with the debriefing session guide 25 min

Figure 1 The components of HS.

Training Method Questionnaire as a post-test. In the second stage, each group of students was individually invited to the simulation laboratory's “Natural Childbirth Room.” They were then asked to practice their natural childbirth

approach skills on a standardized patient. The researcher who evaluated the students' performances was blinded and uninvolved in the HS or VM training. The Natural Birth Approach Clinical Skills Guide was used for evaluation. In

1. Creating a natural birthing environment
2. Providing freedom of movement during labor
3. Providing psychological and physical support during labor and delivery
4. Avoiding unnecessary intervention during labor
5. Intervention for birth pain
6. Applying appropriate massage techniques that support natural childbirth
7. Applying heat or cold to reduce labor pain
8. Using vocalization to reduce labor pain
9. Playing music that the pregnant woman likes to reduce labor pain
10. Using daydreaming to reduce labor pain

Figure 2 The scenario stages.

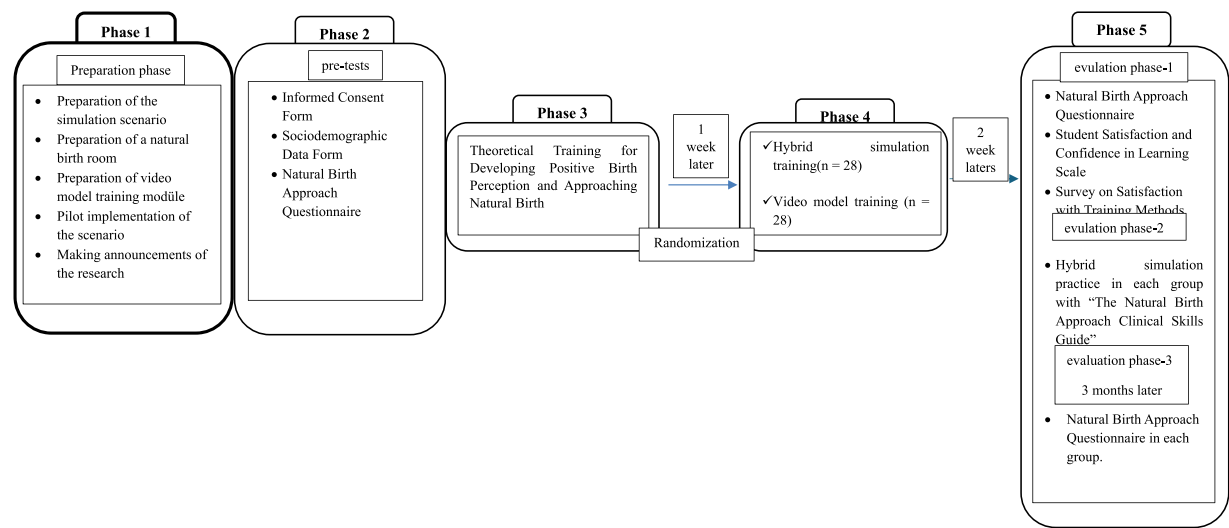


Figure 3 The study protocol.

the final stage, the Natural Approach to Childbirth Questionnaire was readministered to both groups three months after completing the training (Figure 3).

Data collection tools

Sociodemographic Data Form: The form had nine questions that gathered student information, including age, gender, residence (current and during education), high school graduation, reason for choosing nursing faculty, satisfaction with nursing studies, average GPA, and self-perception as a successful student.

Natural Birth Approach Questionnaire: This form gathered information about students’ knowledge, opinions, and approaches to natural birth and consists of eight multiple-choice questions and 29 three-point Likert-type statements. Correct answers in the knowledge test are worth 12.5 points, with scores ranging from 0 to 100 (Güleç Şatır et al., 2018; Ministry of Health, 2018; Lamaze, 2007).

The form underwent a review by ten experts, and necessary revisions were made based on their opinions.

Student Satisfaction and Confidence in Learning Scale: The National League for Nursing developed a 13-item tool to measure students’ attitudes and beliefs toward simulation, with a higher score indicating higher satisfaction and self-confidence levels. The tool consists of two subdimensions and is measured on a five-point Likert scale. The scale’s reported internal consistency coefficient was 0.94 (Franklin et al., 2014; Karaçay & Kaya, 2017), which was found to be 0.95 in this study.

Survey on Satisfaction with Training Methods: The Training Satisfaction Scale measures students’ satisfaction with their training. It has 16 statements and a score range of 16 to 80, with higher scores indicating higher satisfaction. The scale’s Cronbach’s alpha value is 0.90 in this study, and 0.91 indicates good reliability (Gürpınar et al., 2010).

Natural Childbirth Approach Clinical Skills Guide: The researchers developed 21 items to evaluate the care given

to a pregnant woman based on the natural childbirth approach. During the simulation-based educational sessions, the guide was used to evaluate the effectiveness of the training. Each item was scored on a scale of one to three. The maximum score is 63, whereas the minimum score is 21. Before implementation, ten field experts reviewed and provided their expert opinions (Ünsal-Atan et al., 2019; Boz et al., 2019).

Data analysis

Statistical analyses were performed using Statistical Package for Social Sciences 24 (IBM, USA). Descriptive statistical methods were used, including frequency, percentage, mean, standard deviation, median, and variance analysis. The data were analyzed for normally distributed conformity using the Shapiro–Wilk test. As the data did not follow a normal distribution ($p < .05$), intergroup comparisons were performed using the Mann–Whitney U-test. Wilcoxon’s signed rank test was used to compare pre- and poststudy values within the same group. Chi-square analysis was used to evaluate categorical variables. The results were assessed at a 95% confidence interval and significance at $p < .05$ level.

Ethical considerations

This study was approved by the Ege University Health Sciences Scientific Research and Publication Ethics Board (12/04;1161). Participation in the study was voluntary. Participants were provided with an informed consent form, including a written explanation of the purpose, content, expected duration, number of volunteers, location, and participants’ rights. Students who chose not to participate were informed that this would not negatively affect their academic grades or status. Students who did not participate continued their routine clinical rotations or practice. At the end of the research data collection, it was announced that students who did not wish to participate could receive this training within the determined date range.

Results

Table 1 presents the sociodemographic features of participants. Both groups shared similar characteristics. Most participants were females ($n = 46$; 82.14%); the mean age was 23.25 ± 1.43 and 22.35 ± 1.12 years in the HS and VM groups, respectively. Table 2 shows that HS and VM training increased students’ knowledge of natural birth. The pre-test scores for both groups were similar, but the post-test scores of both groups increased without significant difference. Both groups’ pre-test and post-test scores differed significantly ($p < .05$). The appendix provides pre-test, post-test, intragroup, and intergroup comparisons of stu-

dents’ opinions regarding natural birth. Most participants agreed that birth is a natural process that should occur without intervention. They believed that policies supporting natural birth should be developed to regulate hospitals. Moreover, they agreed that not all women required synthetic oxytocin during childbirth and that the natural birth of patients with full-term pregnancies should be prioritized. Participants also believed that initiating labor with natural methods should be preferred and that pregnant women should be left alone as much as possible to activate their oxytocin. Lastly, they believed the delivery room should only include people the pregnant woman trusts and wants to be with. Both groups believed in the need for continuous fetal monitoring. Routine amniotomy, oxytocin infusion, and episiotomy were deemed unnecessary. Most students thought that routine episiotomy was not required, but not doing it could increase the risk of vaginal wall injuries and urinary incontinence. The post-tests revealed an increased number of students who agreed that “hospital regulations should support natural birth.” The number of students who decided that “birth control should always be in the health team” decreased. The study found an increase in the HS group’s participants, but no change in the VM group’s belief that not all women need synthetic oxytocin during childbirth. Furthermore, an increase was observed in the HS group and a decline in the VM group, who think that natural methods should be preferred in overdue pregnancies. In both groups, fewer students believed oral fluid intake during labor required intervention and more believed having birth tables in the lithotomy position hindered the natural birth. The HS group agreed that a woman should feel the pushing sensation before starting the pushing process. They also disagreed more with the statement that routine episiotomy in every delivery protects the pelvic floor. The HS group had more students agreeing that intervention during labor reduces the effects of hormones and that trusted people should be in the delivery room. Conversely, the VM group had fewer students who agreed with these statements. After the post-test, both groups had a decreased number of students who believed in continuous fetal monitoring, routine amniotomy, oxytocin infusion, delivery in lithotomy position, and continuous fluid infusion.

The HS group had fewer people who believed routine enemas were necessary, and more people who believed episiotomy should only be performed as indicated. The relevant data is included in the appendices. Table 3 shows the student satisfaction and self-confidence scale and the satisfaction with the education method scale score averages. The study demonstrated that both groups of students scored above average on the Student Satisfaction and Self-Confidence Scale and the Satisfaction with Education Method Scale. Although the HS group had slightly higher scores than the VM group, the difference was insignificant ($p > .05$). The clinical skill scores for the natural birth approach are presented in Table 4. The HS group

Table 1 – Sociodemographic Characteristics.

	HS Group (n = 28)	VM Group (n = 28)	Test
Age	23.25 ± 1.43	22.35 ± 1.12	$z = 2.59^*$ $p = .012$
Gender (%)			
Female	21 (75)	25 (89.3)	$\chi^2 = 1.948^{**}$
Male	7 (25)	3 (10.7)	$p = .163$
Academic success levels	3.49 ± 0.28	3.51 ± 0.28	$z = -0.264$ $p = .793$
Satisfaction with studying nursing			
Yes	23 (82.1)	21 (75.0)	$\chi^2 = 0.424$
No	5 (17.9)	7 (25.0)	$p = .515$

* Mann-Whitney U test.

** Pearson Chi-Square.

Table 2 – Natural Birth Knowledge Scores.

		HS group X ± SS (Min–Max)	VM group X ± SS (Min–Max)	Test
Natural Birth Knowledge	Pretest	32.14 ± 21.36 (0–75)	31.25 ± 18.78 (12–75)	$z = -0.042^{**}$ $p = .966$
	Post-test	52.67 ± 21.61 (0–100)	51.78 ± 16.56 (25–87.50)	$z = -0.427$ $p = .670$
Test		$z = -3.616^*$ $p = .000$	$z = -3.855$ $p = .000$	

* Wilcoxon Signed Rank Test.

** Mann-Whitney U test.

Table 3 – Student Satisfaction and Self-Confidence Scale and the Satisfaction with Education Method Scale Score Averages.

	HS Group X ± SS (Min–Max)	VM Group X ± SS (Min–Max)	HS Group Mean Rank	VM Group Mean Rank	Test	
					Z*	p
Satisfaction in Learning and Self-Confidence Scale	57.10 ± 7.93 (39–65)	55.35 ± 5.63 (48–65)	30.43	26.57	–0.896	.370
Satisfaction with Learning Subdimension	22.53 ± 3.22 (15–25)	21.92 ± 2.10 (20–25)	31.64	25.36	–1.514	.130
Self-Confidence Subdimension in Learning	30.60 ± 4.39 (21–35)	29.50 ± 3.39 (24–35)	30.73	26.27	–1.043	.297
Satisfaction Survey with Educational Methods	69.25 ± 7.71 (48–80)	68.35 ± 7.47 (58–80)	29.46	27.54	–0.446	.656

* Mann-Whitney U-test.

scored 58.78 ± 3.04 , whereas the VM group scored 40.32 ± 4.75 . The difference between the two groups was found to be significant. The students in the intervention group who received training with hybrid simulation had significantly higher natural birth approach skill scores than those in the VM group ($p < .001$).

Discussion

This study compared nursing students' natural birth knowledge, perception, satisfaction, self-confidence, and natural birth approach skills in the HS and VM groups. The study reported no significant difference in the knowledge

Table 4 – The Natural Birth Approach Skills Results.

Group	Natural Birth Approach Skill Points		Test	
	X ± SS (Min–Max)	Mean Rank	Z*	p
HS	58.78 ± 3.04 (52–63)	42.36	–3.616	.000
VM	40.32 ± 4.75 (30–55)	14.64	–3.855	.000

* Mann–Whitney U-test

levels between the HS and VM groups pre- and post-training. Both groups significantly improved their knowledge scores from the pre-test to the post-test. Both training methods effectively improved the knowledge of natural childbirth. Previous studies reported various simulation techniques can improve competence in the field of maternity nursing (Song & Son, 2013; Nugraha et al., 2018; Uppor et al., 2024). Devi et al. (2021) found that students in the simulation group had significantly higher knowledge of birth care than those receiving traditional education (Devi, 2021). Chang et al. (2024) reported that the group who received labor support training through a game-based scenario had higher learning performance levels than those who received video-based learning (Chang et al., 2024). This study's results align with studies showing that SBE improves the students' clinical skills in managing natural childbirth and their knowledge of primary obstetric care (Salehian et al., 2023; Çalim et al., 2020; Jaya et al., 2020).

The quality of birth care is directly associated with the quality of education that health professionals receive (İsbir et al., 2023). Therefore, hybrid simulation can help nursing students improve their understanding of natural birth knowledge. Students' self-efficacy and self-confidence levels can affect how they perceive childbirth. Therefore, students should be supported in reflecting on their knowledge of their clinical skills and developing autonomy in decision-making processes (Aslantekin-Özçoban et al., 2021). Nomura et al. (2022) demonstrated that obstetric simulations with a high-fidelity patient simulator increased students' self-confidence in understanding the care procedure in labor (Nomura et al., 2023). Uppor et al. (2024) found that nursing students' clinical judgment levels improved when they participated in clinical obstetric simulation scenarios (Uppor et al., 2024). Elmonheim and Hassan (2017) used video films to educate students about natural childbirth before the clinical birth experience as an effective learning strategy that enhances students' satisfaction compared with traditional education (Elmonheim & Hassan, 2017). However, a different study on nursing students in Egypt found that the satisfaction level of the group who received VM training in normal birth management was lower (El saidy Eisaa et al., 2019).

In this study, both groups scored above average on the Student Satisfaction and Self-Confidence Scale and the Satisfaction with Education Method Scale. Although the HS and VM groups were satisfied with their training methods, no significant difference was observed in their scores ($p > .05$). Consequently, the training methods used in this study can be considered to enhance students' self-confidence and satisfaction levels. A natural childbirth skill guide is suggested based on evidence-based recommendations combined with technology-assisted educational methods that can enhance nursing students' self-confidence and satisfaction levels. It is possible to have positive perceptions of birth by adopting humanistic values and demonstrating high competence (Betran et al., 2018; İsbir et al., 2023). Pajohideh et al. (2023) reported that simulation-based experiences may be more effective than clinical experiences in developing natural birth skills (Pajohideh et al., 2023). Afulani et al. (2020) also demonstrated that incorporating respectful maternal care into simulation-based training creates an exceptional opportunity to enhance clinical and interpersonal skills (Afulani et al., 2020). In their systematic review, Lous et al. (2020) concluded that hybrid simulation is efficacious in improving communication skills in obstetrics (Le Lous et al., 2020).

This study indicated the HS group had a higher mean score than the VM group when comparing both groups' natural birth approach skills scores. This difference was statistically significant ($p < .05$). The findings of this study align with the outcomes reported in the existing literature. Post-training, both groups were more likely to agree that birth should be monitored without intervention and that policies supporting natural birth should be developed. Our study also found that students who received training with HS regarding the natural birth process were more likely to support it. Their opinions changed in the direction of waiting for the feeling of pushing, avoiding routine episiotomy, avoiding intervention, and taking the person the pregnant woman trusts into the delivery room, compared with those who received VM training. In the post-tests, students who received training with HS showed more positive changes in their views supporting natural birth compared with the VM group. After HS training, students' attitudes toward not

performing routine enema and episiotomy, avoiding unnecessary interventions during delivery, and allowing trusted individuals to be present in the delivery room became more favorable. In Easter et al.'s (2016) study, the proportion of participants who would recommend vaginal birth increased after the twin vaginal birth simulation (Easter et al., 2016). Lee et al. (2023) reported that healthcare providers did not adequately provide information on risks and alternatives, a crucial component of respectful care. They recommended that the training process emphasizes educating about these risks and alternatives (Lee et al., 2023). An education curriculum that promotes natural and evidence-based practices, rather than seeing CS as a preferred mode of delivery, can shape students.

Limitations

The findings cannot be generalized due to the data collected from a single university's student population. Another limitation is that no measurement tool in the literature measures the natural birth approach. Additionally, the impact of the interaction between SP and students on their performance in the study was not assessed using any measurement tool. The study was limited to eighth-semester undergraduate students, and differences in scenario progression and timing may have impacted the results.

Conclusion

This study found that using HS and VM education for nursing students significantly improved knowledge associated with natural childbirth education. Students in the HS group showed more improvement in natural childbirth skills and had a more positive perception of natural childbirth compared with those in the VM group. Using the HS method to train students on natural birth approach skills is recommended.

Declaration of competing interest

The authors declare no conflict of interest.

CRediT authorship contribution statement

Duygu Güleç Şatır: Writing – review & editing, Writing – original draft, Project administration, Methodology, Data curation, Conceptualization. **Şenay Ünsal Atan:** Writing – original draft, Methodology, Conceptualization. **Oya Kavlak:** Data curation, Conceptualization. **Gül Ertem:** Methodology, Conceptualization. **Hale Sezer:** Conceptualization. **Figen Kazankaya:** Investigation, Con-

ceptualization. **Sümeyye Tetik:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Ruken Yağız Altıntaş:** Investigation, Conceptualization. **Selin Ahsun:** Investigation, Conceptualization.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ecns.2024.101649](https://doi.org/10.1016/j.ecns.2024.101649).

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